22 Show that $\dfrac{7x-14}{x^2+4x-12} \div \dfrac{x-6}{x^3-36x}$ simplifies to ax where a is an integer.

Factor first fraction: $7x-14 = 7(x-2)$,

$x^2+4x-12 = (x-2)(x+6)$.

Factor second denominator: x^3-36x

$= x(x-6)(x+6)$. Change division to multiply:

$7(x-2)/[(x-2)(x+6)] \times x(x-6)(x+6)/(x-6)$

Cancel common factors, leaving $7x$.

4 marks total
Factor first fraction
Factor cubic
Multiply by reciprocal
Simplifies to 7x

(Total for Question 22 is 4 marks)